

CHE 311

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9/10

HW # 4

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1. $\mu = 2 \text{ cP} = 2 \text{ mPa}\cdot\text{s}$ $\rho = 1000 \text{ kg/m}^3$ $h = 0.1 \text{ m}$ $\beta = 60^\circ$ $\dot{V} = 3 \text{ mL/s}$

$$\delta = \sqrt[3]{\frac{\dot{m} \cdot 3\mu}{\rho^2 g h \cos \beta}} = \sqrt[3]{\frac{\dot{V} \rho \cdot 3\mu}{\rho^2 g h \cos \beta}} = \sqrt[3]{\frac{\frac{3 \text{ mL}}{s} \cdot \frac{1 \text{ m}^3}{1000 \text{ mL}} \times 3 \times 2 \text{ mPa}\cdot\text{s} \times \frac{1 \text{ Pa}}{1000 \text{ mPa}}}{1000 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \cdot \cos 60^\circ \cdot 0.1 \text{ m}}}$$

$$= 0.00332 \text{ m}$$

$$\langle v \rangle = \frac{2}{3} \left(\frac{1}{2} \frac{\rho g h \cos \beta}{\mu} \right) \delta^2$$

$$= \frac{1}{3} \frac{1000 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times \cos 60^\circ}{2 \text{ mPa}\cdot\text{s} \times \frac{1 \text{ Pa}}{1000 \text{ mPa}}} \times \frac{1 \text{ kg m/s}^2}{1 \text{ Pa}} (0.00332 \text{ m})^2$$

$$= 9.028 \times 10^{-3} \text{ m/s}$$

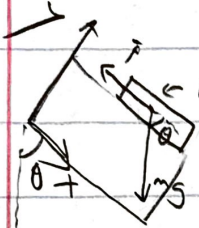
$$Re = \frac{\rho \langle v \rangle L}{\mu} = \frac{\rho \langle v \rangle h}{\mu} = \frac{1000 \text{ kg/m}^3 \times 9.028 \times 10^{-3} \text{ m/s} \times 4 \times 0.00332 \text{ m}}{2 \text{ mPa}\cdot\text{s} \times \frac{1 \text{ Pa}}{1000 \text{ mPa}} \times \frac{1 \text{ kg m/s}^2}{1 \text{ Pa}}}$$

$$= 60$$

Since $20 < Re < 1500$, the flow is laminar

2 $S = 0.1m$ $\beta = 90^\circ - 30^\circ = 60^\circ$ $V_{max} = 0.3m/s$ $\rightarrow V_{xa}$ $\rightarrow V_{xb}$

plank: $\rho = 750 kg/m^3$ $w = 1m$ $L = 1m$ $\delta = 0.05m$ $V = 0.4m/s$



without plank:

$$\int \partial \tau_{yx} = \int \rho g \cos \beta dy$$

$$\tau_{yx} = \rho g \cos \beta y + C_1$$

BC: at $y = \delta$, $\tau_{yx} = 0$

$$C_1 = -\rho g \cos \beta \cdot \delta$$

$$\tau_{yx} = \rho g \cos \beta (y - \delta)$$

$$\tau_{yx} = -\mu \frac{\partial v_x}{\partial y} = \rho g \cos \beta (y - \delta)$$

$$\Rightarrow -\mu v_x = \frac{1}{2} \rho g \cos \beta y^2 - \rho g \cos \beta y \delta + C_2 \Rightarrow -\mu v_x = \frac{\rho g \cos \beta y^2}{2} + \frac{C_2}{\mu} + \frac{(183.9375 - \rho g \cos \beta \delta)}{\mu}$$

BC: $y = 0$, $v_x = 0$

$$v_x = C_2 = 0$$

$$\Rightarrow \mu = -\frac{\rho g \cos \beta y^2}{2 v_{xa}} + \frac{\rho g \cos \beta \delta y}{v_{xa}}$$

$$\Rightarrow \mu = \frac{-\rho g \cos \beta \delta^2}{2 v_{xa}} + \frac{\rho g \cos \beta \delta^2}{v_{xa}} = -\frac{\rho g \cos \beta \delta^2}{2 v_{xb}} - \frac{183.9375 \delta}{v_{xb}} + \frac{\rho g \cos \beta \delta^2}{v_{xb}}$$

$$\rho = -2250 kg/m^3$$

$$\Rightarrow \rho = 2250 kg/m^3$$

$$\mu = \frac{-\rho g \cos \beta \delta^2}{2 v_{xa}} + \frac{\rho g \cos \beta \delta^2}{v_{xa}} = 183.9375 Pa \cdot s$$

with plank:

$$\int \partial \tau_{yx} = \int \rho g \cos \beta dy$$

$$\tau_{yx} = \rho g \cos \beta y + C_3$$

BC: $y = \delta$, $\tau_{yx} = \frac{F}{A} = \frac{\rho V S \cos(60^\circ)}{A}$

$$\tau_{yx} = \frac{\rho V S \cos 60^\circ}{A} = \rho g \cos 60^\circ = 750 kg/m^3 \times 0.05m \times 9.81 m/s^2 \times \cos 60^\circ$$

$$= 183.9375 Pa$$

$$\tau_{yx} = -\mu \frac{\partial v_x}{\partial y} = \rho g \cos \beta y + C_3$$

BC: $y = 0$, $v_x = 0$

$$v_x = 0 = C_4$$

$$\Rightarrow -\mu v_x = \frac{\rho g \cos \beta y^2}{2} + 183.9375 y - \rho g \cos \beta \delta y$$

$$\mu = -\frac{\rho g \cos \beta y^2}{2 v_{xb}} - \frac{183.9375 y}{v_{xb}} + \frac{\rho g \cos \beta \delta y}{v_{xb}}$$

$$= -\frac{\rho g \cos \beta \delta^2}{2 v_{xb}} - \frac{183.9375 \delta}{v_{xb}} + \frac{\rho g \cos \beta \delta^2}{v_{xb}}$$

since direction of force is different from x-direction, so negative sign

